

gheim-ch-pii-212k and gheim-ch-560m: An Open Swiss-Language PII Dataset and NER Model. Technical report.

Joel Barmettler
Independent
Zürich, Switzerland
jbarmettler@proton.me

ABSTRACT

We release gheim-ch-pii-212k, a 212,503-chunk personally-identifiable information (PII) named-entity-recognition dataset covering the four official Swiss languages (Swiss German, Swiss French, Swiss Italian, Romansh) and English, and gheim-ch-560m, a 560M-parameter xlm-roberta-large fine-tune released alongside it. The dataset combines real Swiss text from court rulings, federal parliament records, Swiss-filtered web text and a Romansh corpus with a templated and LLM-generated synthetic gap-fill layer covering cells where the real-text corpus was too sparse for training and evaluation. Annotations come from a three-LLM consensus (Gemma 4 26B-A4B, Qwen3.6 35B-A3B, Nemotron-3 Nano Omni 30B-A3B) plus a regex catalogue with checksum validators, merged with per-span provenance. gheim-ch-560m attains 0.70–0.94 person-NER char F_1 on four external benchmarks (ai4privacy/openpii-1m, ZurichNLP swissner, Babelscape WikiNeural, CoNLL-2003) without ever seeing training data from any of them; on the in-distribution held-out test split it attains a strict-span F_1 of 0.910 (char F_1 0.946). Among open Swiss-applicable detectors run zero-shot on the same Swiss test set, the strongest comparable overall- F_1 entry is Microsoft Presidio at 0.398 strict / 0.501 char. The dataset ships under CC BY 4.0; the model under Apache 2.0; both are available on the Hugging Face Hub. Reproducible eval scripts and checkpoints are on GitHub. This is an engineering report on a publicly-released artifact; no new method or architecture is proposed.

1 INTRODUCTION

The dominant pattern in production large-language-model (LLM) workflows is to call a hosted API: OpenAI, Anthropic, Google, Mistral. For Swiss-market deployments this raises a recurring compliance concern: every request that contains a customer name, a Swiss address, an IBAN, or an AHV/AVS social-security number leaves Swiss jurisdiction the moment it crosses the API boundary. The legal landscape for using third-party LLM services with personal data under Swiss law is analysed in detail by Rosenthal and Veraldi [20]; the practical mitigation in production systems is to anonymise sensitive content before the request and restore the originals on the response. The accuracy of the round-trip depends on the quality of the underlying named-entity recogniser.

Existing public PII detectors handle Swiss text poorly. The closest open generalist, openai/privacy-filter [16] (an OLMoE-based [14] 1.4B-parameter mixture-of-experts model with 50M active parameters), reaches a strict-span F_1 of 0.44 on our held-out Swiss test set despite multilingual training. Microsoft Presidio [13], configured with per-language NLP engines, reaches 0.43. Multilingual NER baselines such as Davlan’s xlm-roberta-base-ner-hrl [1]

reach 0.65 person-only F_1 . None of the available systems fluently handle the four official Swiss languages together with the structured PII formats common in Swiss documents (CH-IBAN, AHV, VAT-CHE), and only one (SwissNER [25]) provides a Romansh evaluation slice.

This report documents three publicly-released artifacts:

- (1) gheim-ch-pii-212k, a multilingual PII NER dataset of 212,503 chunks across Swiss German, Swiss French, Swiss Italian, Romansh and English, covering eight PII categories (account number, address, date, email, person, phone, URL, secret), encoded in a 33-class BIOES tag scheme byte-aligned with openai/privacy-filter.
- (2) gheim-ch-560m, a 560M-parameter token-classification model fine-tuned from xlm-roberta-large [5] on the dataset, attaining a held-out test F_1 of 0.910 strict-span and 0.946 character-level.
- (3) A reproducible evaluation harness that scores the released model on four established external benchmarks (ai4privacy/openpii-1m, ZurichNLP swissner, Babelscape WikiNeural [21], CoNLL-2003 [23]) and seven competing open detectors on the released test set, with replication checks against each baseline’s published numbers on its native dataset.

1.1 Scope

This is a technical report on a deployed artifact, not a research contribution. We do not propose new model architectures, training methods, evaluation metrics, or annotation schemes. The model is xlm-roberta-large [5] fine-tuned with stock AdamW [11] and a small hyperparameter sweep; the label space is byte-identical to openai/privacy-filter [16]; the real-text corpora are taken as-is from the Apertus pretrain release [7] and downstream sources; the dataset construction pattern (LLM annotation followed by checksum-validated regex augmentation and template-based gap-fill) is by now standard practice. The contribution is the assembled artifact: a publicly-released Swiss-language PII dataset that includes Romansh, a strong out-of-the-box checkpoint for the same niche, and a reproducible evaluation that puts both into context against the open alternatives. The report documents what the dataset contains, how the model was trained, what its measured behaviour is on multiple test sets, and what its known limitations are, so downstream users can judge whether to deploy it.

The rest of the paper is structured as follows. Section 2 situates the work against prior art. Section 3 describes the dataset construction. Section 4 covers model selection and training. Section 5 reports the evaluation. Section 6 discusses limitations and Section 7

summarises. Appendix A collects the four full result tables (per-language $\times$ per-category breakdown, comparison to other detectors, cross-domain transfer, and a methodology-validation pass on each external baseline).

2 RELATED WORK

2.1 PII detection and de-identification

Microsoft Presidio [13] is a widely-used industrial open-source PII detector: a per-language spaCy NER engine combined with checksum-validated regex recognisers for structured formats (IBAN, credit card, SSN, etc.). Its built-in catalogue is large and the framework is easy to extend; its weakness on Swiss text is inherited from the per-language spaCy NER backbone (Section 5, Table 6).

The AI4Privacy family of corpora and fine-tuned checkpoints [2, 3] is a high-volume recent open-source effort. Their datasets are templated synthetic prose with embedded Faker-generated PII at the million-chunk scale across 12+ fine-grained categories and 17+ locale variants, and the published distilbert fine-tunes (e.g. `Isotonic/distilbert-finetuned-ai4privacy_v2`) are widely-downloaded open PII detectors on the Hugging Face Hub. Section 5 confirms the published per-token accuracy numbers but shows a sharp drop on real Swiss text: when scored under strict-span F_1 on `gheim-ch-pii-212k` the fine-tunes reach 0.16. The cross-domain gap reflects template-vs-real domain shift more than label-noise: AI4Privacy’s underlying templates were not designed with Swiss surface forms (CH-IBAN, AHV/AVS, +41 phones, German-language addresses) as targets.

`openai/privacy-filter` [16] is the strongest open zero-shot generalist on our test set. It is a token-classification head atop OL-MoE [14], a 1.4B-total-parameter mixture-of-experts encoder with 50M parameters active per token, and it shares our 33-class BIOES schema. Its `openai-namespace` release made it the default upstream from which our schema is byte-aligned, so a fine-tune started from this model can reuse the pretrained classifier-head weights.

2.2 Multilingual NER

XLM-RoBERTa [5] is a common starting point for multilingual token-classification and is the base of `gheim-ch-560m`. The Davlan family of fine-tunes [1] are trained on PER/ORG/LOC across many European languages and provide our generalist baseline. WikiNeural [21] provides large-scale silver-data multilingual NER training and evaluation across nine European languages and is one of our cross-domain external benchmarks. The CoNLL-2003 shared task [23] is a long-running English NER benchmark and is also part of our cross-domain evaluation. The BIOES tag scheme used throughout follows the design analysis of Ratinov and Roth [19].

2.3 Swiss-NLP and the Apertus stack

SwissBERT [24] is an encoder pretrained on Swiss text covering all four official languages, including Romansh. We considered SwissBERT as a base in the bake-off (Section 4, Table 3) and selected XLM-R-large by 0.8 pp validation F_1 ; SwissBERT is a competitive alternative when ~ 300 M parameters is the deployment ceiling. Its companion test set, SwissNER [25], is the only public Swiss-news

NER set that includes Romansh and is one of our cross-domain external benchmarks. The Romansh evaluation in particular relates closely to the WMT24++ benchmark extension of Vamvas et al. [26], which adds Rumantsch Grischun together with the five regional Romansh idioms (Sursilvan, Sutsilvan, Surmiran, Puter, Valader) to the WMT machine-translation benchmark; the Apertus Romansh corpus, which underlies our `rm` chunks, is constructed in a similar idiom-aware way.

The real-text portion of our dataset is sourced from the Apertus pretrain corpora released alongside the Apertus Swiss-language LLM [7]. Apertus aggregates four Swiss-domain sources: the Open Legal Data Platform *Entscheidungsuche*, which provides machine-readable Swiss court decisions across cantons in their original German, French, or Italian [4]; *Curia Vista*, the official business database of the Swiss Federal Assembly, which includes parliamentary motions, postulates, and transcripts in German, French, and Italian [22]; the Swiss-filtered subset of FineWeb-2 [18] (the multilingual web-crawl corpus that scales the FineWeb pipeline of Penedo et al. [17] to all languages and applies XLM-R-based quality filters); and a purpose-built Romansh corpus drawing on cantonal law texts, the Lia Rumantscha online dictionaries, the ZurichNLP bilingual corpus, and Romansh Wikipedia. These corpora are released under CC-BY-4.0-compatible licences and are the substrate for the real-Swiss-text portion of `gheim-ch-pii-212k`.

2.4 Quantisation for browser deployment

The released model ships an int8 ONNX export so the demo at `gheim.ch` runs entirely in-browser via WebGPU or WebAssembly. The quantisation procedure follows the post-training integer-arithmetic-only inference framework of Jacob et al. [10], applied through ONNX Runtime [12] via the optimum dynamic-quantizer with AVX-512 VNNI as the deployment target. The browser runtime is `Transformers.js` [9].

2.5 Industry tooling

`transformers` [27] is the training and inference runtime. `seqeval` [15] provides the strict-span scoring used both for our headline F_1 and for every comparison and replication entry in Section 5. `spaCy` [8] provides the per-language NER pipelines used in our comparison and inside Presidio. `AdamW` [11] is the optimiser. `Faker_CH` provides the Swiss-locale random data used in our synthetic gap-fill payload generation.

3 DATASET: GHEIM-CH-PII-212K

3.1 Source corpora and language coverage

Real-text content is drawn from the Apertus pretrain corpora [7]: the Swiss legal-decisions subset *entscheidungsuche* curated from the Open Legal Data Platform of the same name [4], federal-parliament proceedings drawn from the *curia_vista* business database [22], the Swiss-filtered slice of FineWeb-2 [18], and the Romansh corpus released alongside Apertus, which compiles material from cantonal law texts, the Lia Rumantscha online dictionaries, the ZurichNLP bilingual corpus, and Romansh Wikipedia. Source documents are chunked at sentence boundaries to roughly 1,500 characters, then tagged with `facebook/fasttext-language-identification` at confidence

≥ 0.5 . Chunks below threshold are dropped. The five-language distribution of the released dataset is shown in Table 1.

Table 1: Per-language chunk distribution.

Code	Variant	Chunks	%
de_ch	Swiss German (written std.)	90,292	42.5
fr_ch	Swiss French	64,021	30.1
it_ch	Swiss Italian	36,147	17.0
rm	Romansh	16,514	7.8
en	English (Swiss-context)	5,529	2.6
Total		212,503	100

Spoken-form Swiss German (GSW) is not represented: the corpus is written-standard German as used in Swiss legal and journalistic contexts. The fasttext language identifier labels GSW (“chuchichäschtlī”-style orthography) as standard German, so incidental dialect content cannot be reliably separated.

3.2 Annotation schema

The label space follows the openai/privacy-filter taxonomy: eight PII categories listed in Table 2, each encoded in a BIOES tag scheme [19] (Begin, Inside, Outside, End, Single), giving 33 classes overall (one outside class plus four BIOES positions for each of the eight categories). The byte-identical label mapping was chosen for compatibility: it lets a fine-tune started from openai/privacy-filter’s checkpoint reuse the pretrained classifier-head weights without remapping, and it lets downstream consumers swap the two checkpoints in place. The cost is that the upstream taxonomy’s choices are inherited as-is, including ones we would have made differently. `account_number` lumps four structurally distinct formats (CH-IBAN, AHV/AVS, VAT-CHE, credit cards) into one cell, so the model cannot signal which kind it detected; `secret` is broad enough to catch any high-entropy token that looks key-shaped; there are no `organization` or `location` cells, which some downstream tasks would want. The `private_` prefix is historical; all eight cells are applied with a recall-oriented labelling policy and may flag publicly-listed institutional entities (e.g. a court switch-board phone number).

Table 2: The eight PII categories.

Category	Examples
<code>account_number</code>	CH-IBAN, AHV/AVS, VAT-CHE, credit cards
<code>private_address</code>	Bahnhofstrasse 1, 8001 Zürich
<code>private_date</code>	1990-01-02, 14. März 2026
<code>private_email</code>	alice@example.ch
<code>private_person</code>	Joel Barmettler, Dr. Schmidt
<code>private_phone</code>	+41 44 268 12 34
<code>private_url</code>	https://kanzlei.ch/...
<code>secret</code>	API keys, bearer tokens

3.3 Curation pipeline

Labelling. Real-text content is annotated by three independent open-weights LLM labellers run in parallel on the same inputs: Gemma 4 26B-A4B (4-bit AWQ via vLLM, cyankiwi/gemma-4-26B-A4B-it-AWQ-4bit), Qwen3.6 35B-A3B (4-bit AWQ), and Nemotron-3 Nano Omni 30B-A3B Reasoning (FP8). Each labeller is constrained to the same JSON-schema output so the spans are directly parseable. The first pass used Gemma alone; a small-sample human-eyeball gate ($n=176$) showed $F_1=0.66$ against subagent-resolved gold, with the bulk of the gap in commercial-register director listings and recall-aggressive court-role text. We added Qwen as a second labeller to lift recall on those failure modes, then Nemotron as a third for tie-breaking when Gemma and Qwen disagreed. The three labellers were applied to the full Apertus pre-train corpora (approximately 4.5 million chunks across the five languages); the combined wall-clock cost is roughly 30 GPU-hours on a 2×RTX 4090 setup.

Regex augmentation. A separate pass re-scans the same Apertus corpora with a regex catalogue gated by checksum validators (ISO 13616 mod-97 for IBAN, EAN-13 mod-10 for AHV, ISO 7064 mod-11 for VAT-CHE, Luhn for credit cards). This is an independent scan rather than a post-filter on the LLM output, so it can pick up structured PII in chunks that the LLMs rejected or missed. A column-context heuristic gates the regex hits so that numeric tables (court statistics, parliamentary vote counts) whose digits happen to be IBAN- or AHV-shaped are not mis-flagged as `account_number`; the audit in Section 3.5 shows that the heuristic does not catch every such case, and it is one of the open hardening directions.

Format-aware commercial-register re-labelling. A fourth annotator runs only on the commercial-register subset, with a prompt tuned to that format. This was added after the audit showed the general-purpose labellers under-recalled director and board-member names in long contact-block listings.

Merge with per-span provenance. The four annotator streams (three LLM, one regex) are merged at the span level. Each surviving span carries the tuple of signals that produced it and a confidence score derived from how many sources agreed. Two post-processing steps run unconditionally on the merged output: a Geonames-CH gazetteer demotes spans labelled `private_person` whose surface is a known Swiss place name (e.g. “Chur”, “Sion”) to the appropriate `private_address` sub-span or drops them when the place stands alone, and a label-conflict resolver collapses overlapping spans at the same character offsets, preferring the higher-confidence label. Both steps were added after we observed the earlier pipeline mis-tagged municipality names as people and double-labelled the same characters under two different categories.

Balancing. The merged set of approximately 2.3M candidate chunks is passed through a five-phase balancer to limit document concentration, value memorisation and per-language class imbalance. The cap numbers were chosen from a frequency-analysis pass over the unbalanced output. Phase E drops spans whose provenance-derived confidence is below 0.5 (a span with no signal agreement is unstable enough that we prefer to discard it). Phase

A caps each source document to 30 chunks. Phase B caps each (language, category, normalised value) tuple to 30 chunks (60 for the medium-difficulty Swiss-language person/address/date cells, 100 for any Romansh cell, 200 for English and for synthetic sources), so a single recurring CH-IBAN or person name cannot dominate its cell. Phase C applies per-(language $\times$ source $\times$ category) cell caps tiered by data abundance: up to 24,000 chunks for the data-rich Swiss-language person, address and date cells in legal and parliamentary text; 18,000 for the other major cells; 12,000 for Italian; 10,000 to keep all of English; smaller caps where the corpus simply does not have the material. `secret` is effectively uncapped because it is rare corpus-wide. Phase D injects negative chunks (chunks with no detected PII) at 10 % of positives per language so the model trains on enough out-of-class context. The five phases together bring the candidate set down to the released chunk count.

Synthetic gap-fill. A separate synthetic layer is appended to populate cells where real-Swiss-text coverage was insufficient. Eight templated layers are generated from hand-written fragment files (greetings, signatures, bodies, legal/medical/bank/HR documents, multilingual KYC/HR/bank/ `secret-leak`/doctor-note documents in DE/FR/IT, short-form narrative in all five languages, DE structured layouts (CSV/vCard/YAML), common-word surname positive/negative pairs, adversarial-negative chunks, and a Romansh-context `secret-slot` layer). A composer framework substitutes Faker_CH-generated names, addresses and phones; checksum-validated CH-IBAN, AHV and VAT-CHE; and common secret formats into the templates, then optionally applies one of seven realistic noise overlays (NBSP, broken caps, no-space, markdown, HTML entity, stray punctuation, OCR swap) at 15 % probability per chunk. A slot-fill verifier additionally prompts Gemma to embed pre-generated payload values verbatim into natural multilingual prose for the rarer cells; it accepts a chunk only if every payload value appears as an exact substring in the model output. Synthetic chunks carry a synthetic boolean so consumers can filter to real-only or synthetic-only as required.

3.4 Splits and held-out integrity

Documents are partitioned at the source-document level before any chunk is taken, with stratification by (language $\times$ source), so no chunk in test or validation comes from a document with a chunk in train. Synthetic chunks are placed deterministically (random seed 20260522). Final split sizes are 170,001 / 21,256 / 21,246 chunks for train / validation / test.

3.5 Label-noise audit

The dataset’s labels are machine-generated. No third-party human annotators were contracted, and no subset carries human gold labels. Every model F_1 reported in Section 5 is therefore a measure of agreement between the trained model and its machine-generated labelling source, not a measure of agreement with a human ground truth.

To put a number on the residual disagreement we ran a four-way re-labelling audit. A stratified 580-chunk sample (drawn from the val and test splits, not the training pool) was re-annotated by four named 2026-era LLMs accessed via OpenRouter — moonshotai/kimi-k2.6, deepseek/deepseek-v4-pro,

minimax/minimax-m2.7 and z-ai/glm-5.1 — each given the same per-language prompt the dataset’s own labeller uses, with temperature=0 and seed=0 so the run is reproducible from a checkout. The four per-model labellings are combined into a per-span majority vote (a span is in the consensus iff ≥ 2 of 4 models agree on it). The released dataset’s labels reach an F_1 of **0.71** against that majority vote (precision 0.747, recall 0.680, $n = 580$ chunks, 1,349 gold spans). Per-category breakdown shows the expected spread: structured categories where the surface form carries the signal score high (`private_phone` 0.87, `private_email` 0.86, `private_url` 0.78), free-text categories with policy room score lower (`private_date` 0.72, `private_person` 0.70), and the categories whose definition depends most on the recall-oriented labelling policy score lowest (`account_number` 0.52, `private_address` 0.43; `secret` 0.67 on only 4 spans). Inspection of the `private_person` disagreements shows the gap is dominated by policy interpretation rather than random annotator error: the audit models tend to omit publicly-listed officials, authors of court rulings, and witness names, while the dataset’s recall-oriented policy includes them.

For comparison, the four audit models also disagree among themselves. Per-model F_1 against the dataset’s labels (no voting) ranges from 0.42 (kimi-k2.6, the most conservative) to 0.73 (deepseek-v4-pro, the most policy-aligned). This range bounds what any single audit run could plausibly produce, and motivates the majority-vote consensus as the more stable reference. The raw per-model labellings, the consensus, and the scoring script are committed under `data/audit_openrouter_*.jsonl` so the 0.71 number is reproducible from the released sample.

The 0.71 audit number is best read as a measure of *how aggressively the dataset labels relative to a four-model consensus*, not as a label-noise floor in the random-error sense; the model’s headline 0.910 strict F_1 should be read against this with that caveat. An earlier development pilot ($n=176$, single-model Qwen re-annotation) gave $F_1=0.66$ and was used during dataset construction as a coarse quality gate; the 0.71 number above is the reproducible, four-model successor.

What drives the gap. A natural worry is that the 0.71 F_1 is largely span-boundary noise (commas, titles, the “Dr. Müller” vs “Müller” fragmentation that NER scoring is notoriously sensitive to). It is not. Re-scoring with a fuzzy match that treats any same-label substring overlap as a match yields $F_1=0.73$ — only **1.7 pp** of the gap is boundary noise. The remaining 28 pp falls into three concrete patterns, each pointing at a specific dataset-quality follow-up for a future revision:

- **Regex-layer false positives in `account_number`** (52 dataset-only spans). The regex-mining pass picked up numeric tables in court documents and parliamentary statistics — e.g. “2011 2011 2011 1-3 1-3 1”, “1892 101 14123 1519” — as IBAN/AHV-shaped, but they are not PII. All four audit models reject them. *This is a residual dataset bug a follow-up cleanup pass should tighten further* (the released artifact already applies a column-context heuristic; the audit shows it does not catch every case).
- **Recall-aggressive labelling of court roles** (101 `private_person` dataset-only, 104 `private_address`

dataset-only). The dataset labels witnesses, judges, court reporters, and named places-of-origin in legal opinions; the audit models follow the prompt’s “do not tag generic roles” rule more strictly. This is a real policy choice the dataset makes by design, and a future revision could expose it as an opt-in flag rather than the default.

- **Under-recall of names in commercial-register text** (156 private_person consensus-only). The audit models consistently flag managing directors and board members in Swiss commercial-register listings (e.g. “Reto Rubeli”, “Fabienne Rubeli-Riedo”, “Karim Lassueur”) that the dataset’s labelling pass missed in long contact-block listings. *This is a residual recall gap a follow-up revision could address* by re-labelling the commercial-register subset with a prompt that emphasises this format (the released artifact already runs a format-aware re-labeller on this subset; the audit shows further yield is still on the table).

The artefacts that produced these numbers (raw per-model labellings, consensus, classification breakdown, example chunks) are committed under `data/audit_openrouter_*.jsonl`, `data/audit_openrouter_report.json` and `data/audit_openrouter_analysis.json`, and the scripts that produced them are at `training/src/gheim_training/` `data/analysis/audit_openrouter.py` and `audit_score.py`.

4 MODEL: GHEIM-CH-560M

4.1 Architecture and base-model selection

The model is a token-classification head atop xlm-roberta-large [5] (24 layers, 1024 hidden, 16 heads, 560M parameters). XLM-R-large was selected following a controlled bake-off against ZurichNLP/swissbert [24] (270M dense), summarised in Table 3. Both bases received an identical 5×3 hyperparameter sweep over (learning rate, layer-wise LR decay) at one epoch, after which the winning configuration per base was trained for three full epochs and selected by best validation F₁. A third candidate, openai/privacy-filter, was deferred: the released model is required to ship as an ONNX export so that the in-browser demo at gheim.ch can run via transformers.js, and the Olmo-MoE architecture is not currently stable in optimum’s [12] ONNX export path.

Table 3: Base-model bake-off. Numbers are from the original base-model selection on the predecessor dataset (gheim-ch-pii-171k); the released xlm-roberta-large fine-tune on gheim-ch-pii-212k attains 0.910 val / 0.910 test strict-span F₁. The bake-off has not been re-run on the new dataset because the architecture decision (XLM-R-large over SwissBERT) was settled at the time of selection.

Base model	Val F ₁	Test F ₁
ZurichNLP/swissbert (270M)	0.910	0.910
FacebookAI/xlm-roberta-large (560M)	0.918	0.916

In the (LR, LLRD) sweep, learning rate was the dominant factor: F₁ increased monotonically with LR over the tested range (5e-6 to 5e-5) for both bases. Layer-wise LR decay did not improve any

cell at any tested value (1.0, 0.95, 0.9). The selected configuration is therefore LR = 5×10^{-5} without LLRD.

4.2 Training procedure

The training mix is the train split of the released dataset. An earlier build of the model used an AI4Privacy English/email augmentation slice on top of the in-domain training data to shore up English-anchor coverage and CH-region email recall; the released checkpoint trains only on the in-domain split so the pipeline is end-to-end reproducible from the gheim repository alone. Hyperparameters are listed in Table 4.

Table 4: Training hyperparameters.

Hyperparameter	Value
Optimiser	AdamW [11]
Learning rate	5e-5
LR scheduler	Cosine, 5% warmup
Layer-wise LR decay	1.0 (off)
Effective batch size	128 (per-device 64 × 2 GPUs DDP)
Epochs	3 max; best at step 3,500 of 3,987 (epoch 2.63)
Max sequence length	512
Precision	bf16
Hardware	2 × NVIDIA RTX 4090
Wall time	≈ 66 min train + 5 min eval

The best checkpoint was selected by `eval_overall_f1` on the validation split. The held-out test split was scored exactly once on the selected checkpoint to produce the headline numbers in Section 5.

4.3 Deployment formats and SDKs

The model ships in two formats. Server-side users load `model.safetensors` (fp32 PyTorch, 2.2 GB) via `transformers` [27]; in-browser users load `onnx/model_quantized.onnx` (int8 dynamic-quantised ONNX, 552 MB) via `transformers.js` [9] on WebGPU or WebAssembly. Two thin SDKs wrap the model for the LLM-anonymisation round-trip use case: gheim on PyPI (Python) and gheim on npm (TypeScript / JavaScript). Both expose the same primitives (anonymise text or chat messages, de-anonymise the streamed response, and a session object that preserves the sentinel mapping across calls), and both default to the calibrated decoder discussed in Section 6. A live in-browser demo at gheim.ch runs the ONNX model through the JS SDK end-to-end, with no server-side inference. The int8 ONNX export reproduces fp32 strict F₁ within 0.6 pp on the held-out test split (0.9044 vs 0.9105) and char F₁ within 0.13 pp (0.9448 vs 0.9461), at 557 MB for the quantised file versus 2.2 GB for the fp32 ONNX. Per-category char-F₁ deltas are all within ±0.003 pp; the int8 loss is concentrated almost entirely on private_address (−0.003 pp char), the same cell that is weakest under fp32. Structured-PII categories (account_number, private_email, private_phone, private_url, secret) are unaffected by the int8 step.

5 EVALUATION

5.1 Methodology

Two metrics are reported throughout. **Strict-span F_1** is the standard NER literature metric: a predicted span counts as a true positive only if its (start, end, label) tuple matches a gold span exactly. Computed via seqeval [15] at the token level for HF and ONNX backends, and via char-level set match for span-emitting backends (spaCy, Presidio) that lack a token grid. **Char-level F_1** (label-aware) is computed over (character index, label) pairs: it asks “did the model mask the right characters with the right category?” and is invariant to entity-segmentation policy differences across systems.

The latter matters because cross-model PII evaluation has a structural problem: different annotation conventions split the same surface form differently. ai4privacy/openpii-1m splits a person mention into a GIVENNAME and a SURNAME span; gheim-ch-560m and CoNLL-2003 emit one combined PERSON span; address detectors fragment “Werdstrasse 36, 8004 Zürich” at the comma or do not. Strict-span F_1 penalises every boundary mismatch even when the underlying redaction-utility goal is met. Reporting both metrics lets the reader judge which is appropriate for their use case: strict for literature comparability, char for deployment.

5.2 In-distribution performance

The numbers in this subsection and the next come from a held-out test split drawn from the same Apertus source corpora as the training data, with strict document-level disjointness between splits but the same sampling distribution. They are an in-distribution upper bound rather than a generalization signal; the cross-domain numbers in Section 5.4 are the more meaningful evidence that the model transfers to data drawn from a different distribution.

On the held-out test split of gheim-ch-pii-212k (21,246 chunks), gheim-ch-560m attains a strict-span F_1 of 0.910 (precision 0.890, recall 0.932) and a char F_1 of 0.946. The validation/test F_1 gap is 0.000, consistent with no observable overfitting to the validation set used for checkpoint selection. The full per-language $\times$ per-category char F_1 breakdown is in Table 5 (Appendix A).

Structured categories with strong surface signals (secret, email, phone, URL, account_number) hold above 0.97 char F_1 on average across languages. private_person and private_address carry the per-language variance, with character-level F_1 ranging 0.85–0.96 across the five languages. Romansh is the weakest of the five languages (0.92 char overall, 0.85 char on address) and reflects the smaller Romansh corpus available for training. For production use the account_number and private_address categories are intended to be paired with a deterministic regex front-end that applies checksum validation (see packages/gheim-py/src/gheim/detectors/composite.py).

5.3 Comparison to other systems

Eight other open PII / NER systems were scored on the same test split under the same harness; the full result table is in Table 6 (Appendix A). PER-only models (Davlan, dslim, spaCy) emit only person mentions and no other PII categories, so their overall 8-category F_1 is reported as n/a; the meaningful

number for those is the private_person cell. openai/privacy-filter is scored via its shipped ONNX export because its custom model_type=openai_privacy_filter is not registered in transformers 4.57; the harness patches config.json’s model_type to xlm-roberta so the auto-config loader can read the id-to-label mapping, then runs inference through the unmodified ONNX graph. ZurichNLP/swissbert-ner is run via an XMOD-adaptor-aware wrapper (the model dispatches per chunk’s language) plus a regex fallback for the non-NER categories (IBAN, AHV, VAT-CHE, phone, email, URL, secret).

Of the public detectors we measured on Swiss text in this configuration, both released gheim-ch-560m checkpoints are the strongest on every cell that is directly comparable. The comparison is asymmetric: the gheim checkpoints are in-distribution on this test set (the training data was drawn from the same corpora) and the contestants are all run zero-shot, so the absolute gap is partly a domain-fit effect, not only a model-quality difference. With that caveat noted, the two metrics tell the same ranking story: among the contestants the strongest private_person char F_1 comes from ZurichNLP/swissbert-ner + regex (0.770) and dslim/bert-base-NER [6] (0.834, English slice only); under strict F_1 openai/privacy-filter leads at 0.449. The gap between strict and char is largest for systems whose schema differs most from ours: Isotonic/distilbert_finetuned_ai4privacy_v2 gains +30 pp under char F_1 because its fine-grained schema (FIRSTNAME, LASTNAME, MIDDLENAME, PREFIX, ...) collapses under our 8-category remap and produces highly fragmented spans.

Among the contestants, gheim-ch-560m is also the only entry with native Romansh support; spaCy has no Romansh or English pipelines in this configuration, and dslim/bert-base-NER is English-only. Section 5.4 reports the same model on each contestant’s training distribution for the reverse comparison.

5.4 Cross-domain evaluation

The released model was scored on four established external benchmarks (ai4privacy/openpii-1m, Babelscape WikiNeural, ZurichNLP swissner, CoNLL-2003); the full result table is in Table 7 (Appendix A). For pure-NER datasets (swissner, CoNLL-2003, WikiNeural) only PER maps to a gheim category, so the meaningful number is the PER cell. To anchor the gheim cells, the table reports alongside each row the F_1 of a model originally trained on that benchmark’s distribution (reproduced through our harness).

Row by row, gheim-ch-560m transfers well to three of the four benchmarks despite being run zero-shot and without ever seeing training data from any of them: PER character- F_1 is 0.911 on CoNLL-2003, 0.808 on Babelscape WikiNeural, and 0.938 on AI4Privacy openpii-1m. The notable strict-vs.-char gap on openpii-1m (0.178 PER strict, 0.938 PER char) is a schema artefact: openpii-1m splits each person into separate GIVENNAME and SURNAME gold spans, which fragment into two adjacent private_person spans under our remap even though every PII character is correctly masked.

The cross-domain row that the released gheim-ch-560m checkpoint does *not* carry strongly is Swiss news on swissner: per-language person-character F_1 is 0.54 (de), 0.76 (fr), 0.64 (it), 0.41

(rm). The released checkpoint trains only on the in-domain gheim-ch-pii-212k corpus so that the published pipeline is end-to-end reproducible from this repository alone; the swissner gap is the price of that choice. We ran a controlled multi-source training experiment to quantify the gap and to test whether a commercial-licence-only training mix could close it. Section 5.5 reports the three-variant comparison and the two-checkpoint release decision that follows from it.

5.5 Multi-source training experiment and the two-checkpoint release

The architecture is held fixed (xlm-roberta-large, 560M parameters, same hyperparameters as Table 4) and only the training mix is varied. Three variants are compared:

- **baseline** ($n = 170,001$): the in-domain train split of gheim-ch-pii-212k only.
- **commercial** ($n = 239,001$): baseline + the train splits of ai4privacy/openpii-1m (Apache 2.0, 40k chunks in de/fr/it/en) and unimelb-nlp/wikiann (CC BY 4.0 / ODC-BY over Wikipedia, $\approx 29k$ chunks in de/fr/it/en/rm). All training data is commercially relicensable so the resulting model can ship as Apache 2.0.
- **research** ($n = 253,597$): baseline + the train splits of ai4privacy/openpii-1m (40k) + Babelscape WikiNeural (CC BY-NC-SA 4.0, $\approx 39k$ chunks) + CoNLL-2003 (Reuters research-only, 4.4k). The most restrictive upstream licence binds the released checkpoint, so this variant is non-commercial / research-only.

Validation and test splits are held constant across all three runs (the in-domain val/test of gheim-ch-pii-212k) so the in-distribution numbers are directly comparable. External benchmarks are scored zero-shot on each checkpoint. The full result is in Table 8 (Appendix A.4).

In-distribution, the three checkpoints are statistically indistinguishable (strict-span $F_1 = 0.910 / 0.912 / 0.912$; char $F_1 = 0.946$ across all three). Out-of-distribution, the picture is asymmetric:

- On swissner (Swiss-news NER, zero-shot for all three variants — there is no swissner train split), the commercial mix lifts private_person char F_1 from 0.70 (baseline) to 0.84 (+14 pp); research lifts it to 0.90 (+20 pp). The commercial gain is real but recovers only about half of the research gain.
- On openpii-1m and open-pii-500k (PII benchmarks that share schema family with our label space), both commercial and research lift character-level F_1 into the 0.95–1.0 range; baseline stays at 0.87–0.90.
- On CoNLL-2003 and WikiNeural (news / Wikipedia NER), both the commercial and the research variants regress against the in-domain-only baseline on the private_person cell: the commercial mix loses –36 pp on CoNLL and –13 pp on WikiNeural; the research mix loses –15 pp on CoNLL and –1 pp on WikiNeural, even though it trains on the train splits of both. The cause is the same in both cases — the broader 8-category output schema of the gheim checkpoints produces non-PER false positives

on news / Wikipedia text (where any IBAN-shaped, email-shaped, or URL-shaped string is flagged) that the narrower in-domain-only baseline does not produce. The result is a small but real precision tradeoff for the wider category coverage these variants target.

- On gretelai/synthetic_pii_finance_multilingual (multilingual financial-document PII, Apache 2.0), all three checkpoints converge near 0.60 character- F_1 . No training mix contains financial-document text; the benchmark therefore measures the architecture’s structural-PII transfer rather than mix-induced differences. The flat result motivates adding this dataset to a future commercial training mix.

The conclusion of the experiment is that no single training mix strictly dominates: the commercial mix improves swissner at the cost of CoNLL-2003 / WikiNeural; the research mix improves swissner and openpii cleanly but is non-commercial; the in-domain baseline ships under the cleanest licence but loses on Swiss-news transfer. We therefore publish two checkpoints from this paper:

- joelbarmettler/gheim-ch-560m — the in-domain baseline, released under **Apache 2.0** for production and commercial deployments. Cross-domain swissner PER char F_1 0.70.
- joelbarmettler/gheim-ch-560m-research — the multi-source research variant, released under **CC BY-NC-SA 4.0** with a Reuters research-only rider (from CoNLL-2003), for academic research / benchmarks / non-commercial evaluation. Cross-domain swissner PER char F_1 0.89.

The commercial-mix variant is not published as a separate checkpoint — it does not strictly dominate the baseline on cross-domain, and the half-recovery on swissner does not justify a third release. Closing the commercial swissner gap further will require adding more news-domain, commercially-relicensable NER training data (candidates: MultiCoNER v2 under CC BY 4.0, GermEval 2014 under CC BY-SA 4.0, gretelai/synthetic_pii_finance_multilingual under Apache 2.0); we leave that to a future iteration.

A methodology check that replicates each baseline on its own training-distribution dataset and compares the reproduction to its published number is reported in Appendix A (Table 9). Three of four reproductions match within ± 0.03 ; the fourth (Isotonic) is a metric mismatch (per-token accuracy versus strict-span F_1) rather than a harness bug.

6 LIMITATIONS

- **Recall-oriented labelling policy.** Public Swiss documents contain extensive personally-identifiable information that is already public (court rulings, parliament records, public officials). The dataset labels these as PII and the model learns to flag them. Applications that need to distinguish public from private entities should layer a downstream public-vs-private classifier.
- **private_address test strict F_1 is 0.84** (char F_1 0.92). Boundary placement on multi-token addresses is the dominant error mode.

- **account_number test strict F_1 is 0.99** in the headline, but a small fraction of regex-shaped non-PII (numeric tables in court documents and parliamentary statistics) still slips through. Production deployments should pair the model with a regex front-end with checksum validation (IBAN, AHV, VAT-CHE, Luhn).
- **Romansh test strict F_1 is 0.89** (char F_1 0.93), the weakest of the five languages. The Romansh training material is dominated by a single literary/journalistic register (Rumantsch Grischun), so dialectal or technical RM text is unmeasured.
- **Cross-domain Swiss-news transfer is meaningfully weaker than the in-distribution headline**, with swissner per-language person-character F_1 dropping to 0.54 (de) / 0.76 (fr) / 0.64 (it) / 0.41 (rm). The released checkpoint trains only on the in-domain corpus; an alternative training mix that includes external public NER corpora is the planned next iteration.
- **Swiss German dialect (GSW) is not measured.** The fast-text detector used in data preparation labels GSW as standard German.
- **Annotations are machine-generated.** The dataset was not human-annotated; the four-LLM OpenRouter audit in Section 3.5 (F_1 0.71 overall against the majority-of-four consensus, with policy-interpretation disagreements dominating the gap) is the only quantitative labelling-policy reference we can quote.
- **Re-identification is not in scope.** The model is intended for redaction. It is not designed to identify or link entities to real persons and does not return entity-linked identifiers.

Inference-time suppression, browser quantisation, and the calibrated decoder. A failure mode visible only on deploy probes, not in the test-set metrics: on a small fraction of inputs the raw model emits zero non-O tokens for an entire chunk, even when the chunk contains clear PII. The pattern reproduces consistently when a low-confidence person mention is immediately followed by a structurally distinctive token (e.g. a CH-IBAN), which appears to nudge the per-token argmax over the entire span toward the dominant O class. The released SDKs ship a thin wrapper that subtracts a small constant bias from the O-class logit before argmax (CalibratedDetector, default bias 0.5). On a 212-case forensic edge-case probe combined with a 300-chunk in-distribution test slice, a sweep over {0.0, 0.25, 0.5, 0.75, 1.0, 1.5} identifies bias 0.5 as the Pareto-clean point on both the fp32 PyTorch backend and the int8 quantised ONNX deployment target.

On fp32 the bias has essentially no effect on the probe (91.5% perfect at both 0.0 and 0.5) and gains 0.27 pp char F_1 on the slice at zero cost. The dataset pipeline described in Section 3 (gazetteer-driven address labelling, overlap resolution, multi-LLM consensus) appears to have absorbed most of the original suppression pathology that motivated the wrapper; the wrapper is retained as the SDK default because it remains net-positive rather than because it is doing substantial pathology repair. The more interesting finding concerns the int8 browser deployment (Section 4.3). On the same probe, the int8 model drops from 91.5% perfect at bias 0.0 to 73.1%

perfect at bias 0.0 — an 18.4 pp gap that does not appear in the in-distribution test-set numbers reported in Section 4.3, where int8 costs only 0.13 pp char F_1 . Quantisation appears to perturb the 0-versus-entity margin specifically on long-tail edge cases while leaving mainstream chunks essentially unaffected. The calibrated decoder partially recovers this gap: on int8 it lifts the probe perfect-rate by 1.4 pp (73.1% $\rightarrow$ 74.5%) and slice char F_1 by 0.43 pp, roughly 3 $\times$ the fp32 lift. Browser users therefore see a larger calibration win than server-side fp32 users; neither backend incurs a cost at bias 0.5, and higher bias values (1.0–1.5) trade modest probe gains for slice F_1 losses and are not recommended. The wrapper is the default detector in both the Python and JavaScript SDKs.

7 SUMMARY

This report described gheim-ch-pii-212k and gheim-ch-560m, a Swiss-tuned PII NER dataset and model targeting LLM-API anonymisation round-tripping, a deployment context that existing open detectors handle poorly on Swiss text. On the in-distribution test split both released checkpoints attain a strict-span F_1 of 0.91 (char F_1 0.95); among the public detectors we measured zero-shot on the same set the strongest overall is Microsoft Presidio at 0.398 strict / 0.501 char. On out-of-distribution Swiss news (ZurichNLP/swissner, the only Swiss-news NER benchmark that includes Romansh), the Apache-2.0 baseline attains private_person character F_1 of 0.70 and the research variant attains 0.90 — a 20 pp gain from broader training data even though swissner itself is not used at training time (no swissner train split exists; the gain comes from generalisation off the additional openpii-1m / WikiNeural / CoNLL-2003 supervision in the research mix). The dataset is released under CC BY 4.0; the model under Apache 2.0; both are on the Hugging Face Hub. The full evaluation harness, the per-system JSON outputs, and the reproducible build of the comparison tables are at <https://github.com/joelbarmettler/uzh-gheim>.

Known follow-on work. The dataset has no coverage of spoken Swiss German (the fasttext language identifier labels GSW as standard German) and only one Romansh register (Rumantsch Grischun); both gaps would require additional source corpora rather than pipeline changes. The model’s released checkpoint is 560M parameters, larger than is convenient for edge or browser deployment; a smaller distilled variant would be useful but is not part of this release. The dataset’s recall-oriented labelling policy flags publicly-listed institutional entities (e.g. a court switchboard phone) as PII; applications that require stricter precision would benefit from a downstream public-vs-private entity classifier layered on top. The label-noise audit in Section 3.5 also pointed at three concrete dataset-cleanup directions for a future revision: (a) tightening the regex-layer account_number column-context heuristic to catch the residual false positives in numeric tables, (b) optionally exposing the recall-aggressive policy on court-document roles as an opt-in flag rather than the default, and (c) further hardening the commercial-register re-labelling pass so director and board-member names stop being under-recalled in long contact-block listings.

REFERENCES

- [1] David Ifeoluwa Adelani. 2022. XLM-RoBERTa-base fine-tuned on multilingual NER (high-resource languages). Hugging Face model. <https://huggingface.co/Davlan/xlm-roberta-base-ner-hrl>
- [2] AI4Privacy. 2024. PII Masking 200k: An English-language PII dataset. Hugging Face dataset. <https://huggingface.co/datasets/ai4privacy/pii-masking-200k>
- [3] AI4Privacy. 2024. PII Masking OpenPII-1M: A multilingual PII dataset. Hugging Face dataset. <https://huggingface.co/datasets/ai4privacy/pii-masking-openpii-1m>
- [4] Association Entscheidsuche. 2025. Entscheidsuche.ch: Open Legal Data Platform for Swiss court decisions. <https://entscheidsuche.ch/docs>. Accessed: 2025-08-31.
- [5] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised Cross-lingual Representation Learning at Scale. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (2020), 8440–8451. <https://doi.org/10.18653/v1/2020.acl-main.747>
- [6] Sasha Dieguez. 2020. BERT Base NER: A BERT model fine-tuned on CoNLL-2003. Hugging Face model. <https://huggingface.co/dslim/bert-base-NER>
- [7] Alejandro Hernández-Cano, Alexander Hägele, Allen Hao Huang, Angelika Romanou, Antoni-Joan Solergibert, et al. 2025. Apertus: Democratizing Open and Compliant LLMs for Global Language Environments. [arXiv:2509.14233](https://arxiv.org/abs/2509.14233) <https://arxiv.org/abs/2509.14233>
- [8] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. spaCy: Industrial-strength Natural Language Processing in Python. <https://doi.org/10.5281/zenodo.1212303>
- [9] Hugging Face. 2024. Transformers.js: Run Transformers directly in the browser. <https://huggingface.co/docs/transformers.js>
- [10] Benoit Jacob, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. 2018. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. [arXiv:1712.05877](https://arxiv.org/abs/1712.05877) <https://arxiv.org/abs/1712.05877>
- [11] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. [arXiv:1711.05101](https://arxiv.org/abs/1711.05101) <https://arxiv.org/abs/1711.05101>
- [12] Microsoft. 2018. ONNX Runtime: A cross-platform inference and training accelerator. <https://onnxruntime.ai>
- [13] Microsoft. 2023. Presidio: Context-aware, pluggable and customizable PII anonymization service. <https://github.com/microsoft/presidio>
- [14] Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, Yizhong Wang, Akshita Bhagia, Mengzhou Liu, Ben Petrov, Alana Tsai, Hannaneh Hajishirzi, Noah A. Smith, Luke Zettlemoyer, Iz Beltagy, Dirk Groeneveld, Luca Soldaini, and Hannaneh Hajishirzi. 2024. OLMoE: Open Mixture-of-Experts Language Models. [arXiv:2409.02060](https://arxiv.org/abs/2409.02060) <https://arxiv.org/abs/2409.02060>
- [15] Hiroki Nakayama. 2018. sequeval: A Python framework for sequence labeling evaluation. <https://github.com/chakki-works/sequeval>
- [16] OpenAI. 2025. Privacy Filter: A token-classification model for PII detection. Hugging Face model. <https://huggingface.co/openai/privacy-filter>
- [17] Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. 2024. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. [arXiv:2406.17557](https://arxiv.org/abs/2406.17557) <https://arxiv.org/abs/2406.17557>
- [18] Guilherme Penedo, Hynek Kydlíček, Vinko Sabolčec, Bettina Messmer, Negar Foroutan, Amir Hossein Kargaran, Colin Raffel, Martin Jaggi, Leandro Von Werra, and Thomas Wolf. 2025. FineWeb2: One Pipeline to Scale Them All – Adapting Pre-Training Data Processing to Every Language. *arXiv preprint arXiv:2506.20920* (2025). <https://arxiv.org/abs/2506.20920>
- [19] Lev Ratinov and Dan Roth. 2009. Design Challenges and Misconceptions in Named Entity Recognition. In *Proceedings of the Thirteenth Conference on Computational Natural Language Learning (CoNLL-2009)*. 147–155.
- [20] David Rosenthal and Livio Veraldi. 2025. Training AI language models with third-party content and data from a legal perspective. *Jusletter-IT* (March 2025). <https://doi.org/10.38023/bec9257a-a6bb-41cf-b2f3-fda8ae3b448d>
- [21] Simone Tedeschi, Valentino Maiorca, Niccolò Campolungo, Francesco Cecconi, and Roberto Navigli. 2021. WikiNeuRal: Combined Neural and Knowledge-based Silver Data Creation for Multilingual NER. In *Findings of the Association for Computational Linguistics: EMNLP 2021*. 2521–2533. <https://doi.org/10.18653/v1/2021.findings-emnlp.215>
- [22] The Swiss Parliament. 2024. Curia Vista: business database of the Swiss Federal Assembly. <https://www.parlament.ch/en/ratsbetrieb/curia-vista>. Accessed: 2024-04-24.
- [23] Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition. In *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*. 142–147.
- [24] Jannis Vamvas, Noëmi Aepli, and Rico Sennrich. 2023. SwissBERT: The Multilingual Language Model for Switzerland. [arXiv:2303.13310](https://arxiv.org/abs/2303.13310) <https://arxiv.org/abs/2303.13310>
- [25] Jannis Vamvas, Noëmi Aepli, and Rico Sennrich. 2023. SwissNER: A NER test set for Swiss Standard German, French, Italian, and Romansh. Hugging Face dataset. <https://huggingface.co/datasets/ZurichNLP/swissner>
- [26] Jannis Vamvas, Ignacio Pérez Prat, Not Battista Soliva, others, and Rico Sennrich. 2025. Expanding the WMT24++ Benchmark with Rumantsch Grischun, Sursilvan, Sutsilvan, Surmisan, Puter, and Vallader. [arXiv:2509.03148](https://arxiv.org/abs/2509.03148) <https://arxiv.org/abs/2509.03148>
- [27] Thomas Wolf, Lysandre Debut, Victor Sanh, et al. 2020. Transformers: State-of-the-Art Natural Language Processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. 38–45. <https://doi.org/10.18653/v1/2020.emnlp-demos.6>

A DETAILED EVALUATION RESULTS

This appendix contains the four full result tables referenced from Section 5: the in-distribution per-language $\times$ per-category breakdown for gheim-ch-560m; the comparison to seven other public detectors on the same test split; the cross-domain transfer numbers on four external benchmarks; and a methodology check that re-runs each external baseline on its own training distribution. The prose in Section 5 summarises what these tables show; the tables themselves are kept out of the main narrative so each can be inspected at full width.

A.1 In-distribution per-language $\times$ per-category

Table 5 gives the per-language $\times$ per-category char-level F_1 on the held-out test split of gheim-ch-pii-212k. Structured categories with strong surface signals (secret, email, phone, URL) hold above 0.95 in every language; account_number carries most of the per-language variance, with the Romansh cell as the only one below 0.5.

A.2 Comparison to other detectors

Table 6 reports seven other open PII / NER systems on the gheim-ch-pii-212k test split under the same harness. Both the overall 8-category metric and the private_person cell are shown for each system; PER-only systems have n/a for the overall metric since they cannot emit the other seven categories. The comparison is in-distribution for gheim-ch-560m and zero-shot for the seven contestants; see Section 5.4 for the reverse direction.

A.3 Cross-domain transfer

Table 7 reports gheim-ch-560m PER F_1 on four external benchmarks the model has never seen. To anchor each row, the same column reports a model trained on (or claimed-state-of-the-art on) that benchmark’s native distribution, reproduced through our harness. The footnotes give the specific metric / slice each native number is computed on.

[†]Isotonic’s strict-span F_1 on its own training distribution; per-token accuracy across all categories (the metric Isotonic itself reports) is 0.969 in our reproduction versus 0.955 claimed. [‡]Davlan-XLMR’s overall multilingual avg from our reproduction (claimed ~ 0.84). [§]spaCy de_core’s overall F_1 on the German slice of swissner (spaCy ships per-language pipelines and has no single multilingual model that covers the whole set).

Table 5: Per-language $\times$ per-category char-level F_1 on the held-out test split of gheim-ch-pii-212k under gheim-ch-560m. The right-most column gives the per-category average (over languages, gold-weighted); the bottom row gives the per-language average (over categories).

Category	de_ch	fr_ch	it_ch	rm	en	Avg.
account_number	0.994	0.998	0.990	0.971	1.000	0.994
private_address	0.933	0.911	0.916	0.853	0.955	0.917
private_date	0.951	0.908	0.952	0.919	0.883	0.933
private_email	0.996	1.000	0.997	0.989	1.000	0.997
private_person	0.913	0.939	0.951	0.909	0.955	0.930
private_phone	0.995	0.996	0.996	0.985	1.000	0.995
private_url	0.990	0.993	0.993	0.994	0.970	0.991
secret	0.994	0.999	1.000	0.999	n/a	0.997
Avg.	0.944	0.931	0.956	0.918	0.954	0.940

Table 6: Other models on the gheim-ch-pii-212k test split. Strict and char F_1 for both the overall 8-category metric and the private_person cell.

Model	Strict F_1	Char F_1	PER strict	PER char
joelbarmettler/gheim-ch-560m	0.910	0.946	0.896	0.929
joelbarmettler/gheim-ch-560m-research	0.912	0.946	0.898	0.929
dslim/bert-base-NER (English slice)	n/a	n/a	0.561	0.834
ZurichNLP/swissbert-ner + regex hybrid	0.346	0.613	0.464	0.770
Davlan/distilbert-base-multilingual-cased-ner-hrl	n/a	n/a	0.569	0.758
Davlan/xlm-roberta-base-ner-hrl	n/a	n/a	0.610	0.733
openai/privacy-filter (1.4B MoE, ONNX, zero-shot)	0.449	0.619	0.436	0.582
spaCy de/fr/it_core_news_lg	n/a	n/a	0.473	0.574
Microsoft Presidio (multi-lang config)	0.398	0.501	0.433	0.519
Isotonic/distilbert_finetuned_ai4privacy_v2	0.140	0.441	0.243	0.513

Table 7: gheim-ch-560m PER F_1 on four external benchmarks, anchored against a specialist model trained or claimed-state-of-the-art on each benchmark’s native distribution. gheim-ch-560m is run zero-shot; native baselines are reproduced through the same harness.

External benchmark	n	gheim PER strict	gheim PER char	Native baseline	Native F_1
ai4privacy/pii-masking-openpii-1m	8,000	0.178	0.938	Isotonic/distilbert_ai4privacy	0.78 [†]
ai4privacy/open-pii-masking-500k	8,000	0.540	0.933	— (sibling of openpii-1m)	—
gretelai/synthetic_pii_finance_multilingual	4,800	0.519	0.624	— (no public baseline)	—
Babelscape WikiNeural	8,000	0.760	0.808	Davlan/xlm-r-base-ner-hrl	0.813 [‡]
ZurichNLP swissner	800	0.637	0.702	spaCy de_core_news_lg	0.841 [§]
CoNLL-2003	3,453	0.854	0.911	dslim/bert-base-NER [6]	0.957

A.4 Multi-source training experiment

Table 8 reports the full three-variant comparison summarised in Section 5.5. In-domain numbers are character-level F_1 on the held-out test split of gheim-ch-pii-212k; cross-domain numbers are PER character-level F_1 on each external benchmark.

[↓]The commercial variant regresses against baseline on these two benchmarks because the substitute unimelb-nlp/wikiann training data is short Wikipedia spans, lower per-chunk signal density than WikiNeural-train or CoNLL-train; the research variant includes both as training material and so does not regress.

A.5 Methodology validation: replicating each baseline on its own training distribution

To check that the harness in Section 5 is sound rather than an unfair implementation, we replicated each external system on the dataset on which it was trained and compared our reproduction to the system’s own published F_1 (Table 9). Three of four reproductions match within ± 0.03 ; the fourth (Isotonic) is a metric mismatch discussed below.

[†]Isotonic’s claimed F_1 of 0.955 initially reproduced as 0.78 strict-span F_1 on the same data, an apparent 17-pp gap. The difference traces to the metric: Isotonic’s reported number is *per-token classification accuracy* (which includes the dominant 0 class and is much higher than per-span F_1). Reproducing Isotonic’s metric exactly (per-token accuracy including 0) yields 0.969, matching their 0.955 within sampling noise. Our strict-span F_1 of 0.78 on the same data is the correct NER-literature comparison.

Table 8: Three training-mix variants compared. “Baseline” trains on in-domain data only and ships as gheim-ch-560m (Apache 2.0). “Research” adds three external NER / PII corpora and ships as gheim-ch-560m-research (CC BY-NC-SA 4.0 with a Reuters research-only rider). “Commercial” (openpii + WikiAnn only) is reported as an experimental data point and is not separately released.

	Baseline (Apache 2.0)	Commercial (experimental)	Research (CC BY-NC-SA)
Train chunks	170,001	239,001	253,597
External training corpora	—	openpii + WikiAnn	openpii + WikiNeural + CoNLL-2003
In-domain test strict F_1	0.910	0.912	0.912
In-domain test char F_1	0.946	0.946	0.946
<i>private_person-cell character-level F_1 on each external benchmark</i>			
ZurichNLP/swissner (zero-shot)	0.702	0.844	0.903
de	0.539	0.805	0.931
fr	0.761	0.845	0.913
it	0.643	0.745	0.856
rm	0.409	0.552	0.873
ai4privacy/openpii-1m	0.938	0.995	0.995
ai4privacy/open-pii-500k	0.933	0.973	0.982
gretelai/synthetic_pii_finance	0.624	0.619	0.627
Babelscape WikiNeural	0.808	0.681 [↓]	0.795 [↓]
CoNLL-2003	0.911	0.548 [↓]	0.765 [↓]

Table 9: Each baseline scored on its own training-distribution dataset, against published numbers.

Replication target	Claimed	Repro	Δ
dslim/bert-base-NER on CoNLL-2003	0.910	0.913	+0.003
dslim PER on CoNLL-2003	~0.96	0.957	−0.003
Davlan-XLMR on WikiNeural avg	~0.84	0.813	−0.027
spaCy de_core on swissner_de	~0.83	0.841	+0.011
Isotonic distilbert on AI4Privacy	0.955	0.969 [†]	+0.014